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RESEARCH ARTICLE

Inference-Time Noise Addition for Improving Adversarial Robustness of Audio Deepfake Detection System

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ABSTRACT As artificial intelligence technology advances, the generalization performance of audio deepfake detection systems has improved rapidly and significantly. However, these systems still face substantial challenges when confronted with adversarial attacks. To address this issue, many studies have proposed defense methods against such attacks. Nevertheless, only a limited number of studies have focused on defenses in the context of audio deepfake detection. With the development of deep learning, deepfake detection methods are increasingly designed and deployed based on deep learning models. As a result, deepfake detection systems can be applied across various fields and are not restricted to specific data types or domains. Such approaches are referred to as ‘domain-independent’ defense methods and can be applied to a wide range of tasks, including adversarial training. However, they have limitations, such as requiring significant computing resources or potentially degrading original performance. In this paper, we propose Inference-Time Noise Addition (ITNA), an effective and efficient adversarial attack defense method that minimizes the original performance degradation. ITNA adds Gaussian noise once to the input audio sample during the inference phase. It is highly resource efficient, as it does not require additional training or separate AI components. Furthermore, ITNA minimizes performance degradation as much as possible and provides scalability that can be used with other defense methods. We provide a theoretical explanation of the effectiveness of ITNA and validate it through experimental results. Our experiments were conducted on the ASVspoof2021 DF evaluation (official deepfake audio dataset from the ASVspoof community), In-The-Wild (collected in real environments) and DSD-Corpus datasets (includes various synthesizer samples). And experiments results demonstrate the effectiveness of ITNA, reducing the average misclassification rate and the attack success rate of detection systems by 4.48% and 31.90%, respectively. We also demonstrate the effectiveness of ITNA through comparative experiments with alternative noise type and existing noise-based defense method.

INDEX TERMS Adversarial attack defense, audio deepfake detection, inference-time noise addition.

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I. INTRODUCTION

Recent advancements in deep learning have significantly accelerated the development of speech synthesis technologies, including Text-to-Speech (TTS) and Voice Conversion

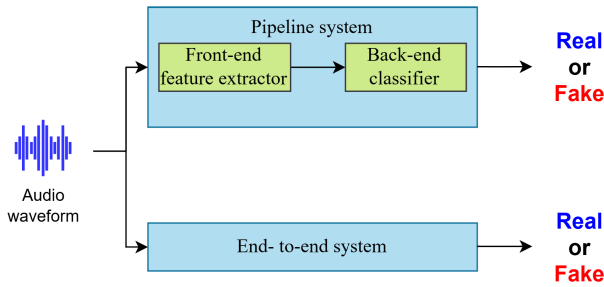


FIGURE 1. Audio deepfake detection workflow: Pipeline system and end-to-end system.

(VC) [1]. In particular, deep generative models such as Generative Adversarial Networks (GANs) [2] and diffusion models [3] have enabled the creation of highly realistic synthetic speech. These audio deepfake [4] technologies offer substantial benefits in various fields [5], such as multilingual narration for social media [6], AI-powered virtual assistants [7], and AI-generated music covers. Their applications continue to expand across domains like healthcare [8], education [9], and customer service, enhancing user experiences and increasing accessibility.

On the other hand, the misuse of increasingly natural-sounding audio deepfakes poses serious problems. With the widespread availability of social media and call recordings, it has become easy to collect other people's voices. Using advanced deepfake audio synthesis models [10], [11], [12], [13], it is now possible to generate highly similar synthetic speech with only a small amount of audio data. As a result, crimes such as voice phishing, impersonation, and fraud using audio deepfakes have been on the rise.

To counter these risks, various audio deepfake detection methods have been developed [14], [15], [16]. As shown in Figure 1, audio deepfake detection systems typically have two workflows. The first is a pipeline-based approach that relies on handcrafted features. The second is an end-to-end approach that uses deep learning to directly extract representations from raw audio for detection. Although end-to-end systems offer better generalization and detection accuracy [17], they are also more susceptible to adversarial attacks due to their deep neural architecture [18], [19].

Adversarial attacks [20] exploit vulnerabilities in neural models by introducing small, imperceptible perturbations to the input waveform, as shown in Figure 2. These perturbations can mislead the detection system into incorrect classifications, undermining its reliability [21]. Attacks are commonly categorized as white-box or black-box, depending on the attacker's access to model information. Despite advances in audio deepfake detection, they remain highly vulnerable to both types of attack.

White-box attacks use all information about the target model, such as gradients, weights, input and output. In particular, gradients and weights reflect the model's learned parameters and how it processes input data, enabling attacks that are powerful enough to achieve nearly 100% success rates.

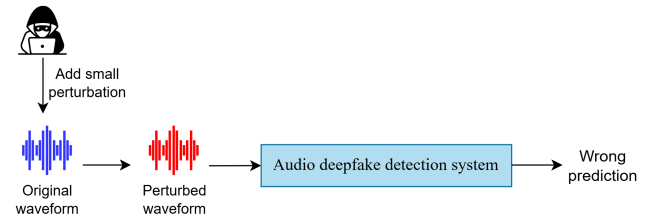


FIGURE 2. Adversarial attack process in audio deepfake detection system.

Black-box attacks only use the input and output information of the model. Due to limited information about the target model, researchers are exploring various attack strategies, including query-based and transfer-based attacks [22], [23], [24].

In this way, attackers can deceive the system using either full internal model access or merely observable input and output behavior. Therefore, even if a model is designed with good generalization and robustness in mind, it cannot be said to be completely safe from adversarial attacks. Furthermore, the opaque and highly integrated nature of end-to-end models makes it difficult to identify and mitigate specific vulnerabilities. This highlights the need for effective and practical defense mechanisms, especially those that do not rely on retraining or modifying existing model architectures. In the context of audio deepfake detection, these challenges are even more critical. Unlike image data, audio signals are temporally continuous, perceptually sensitive, and difficult to process without introducing noticeable distortion. Existing defense methods require high computational costs [25], retraining [26], and the introduction of additional modules [27]. Furthermore, the use of neural networks in this domain is significantly constrained, as even minor artifacts introduced by defense mechanisms can severely degrade detection performance.

In this paper, we propose Inference-Time Noise Addition (ITNA), a simple yet effective defense method that adds Gaussian noise into audio inputs during inference. This approach mitigates adversarial attacks without requiring model retraining. We examine the characteristics of adversarial examples and perturbations, and analyze the impact of noise addition. We theoretically demonstrate the effectiveness of ITNA in defending against adversarial attacks. In addition, we validate the effectiveness of ITNA through experiments on two structurally different models and three datasets. The experimental results show that ITNA is effective against both white-box attacks and transfer-based black-box attacks. During this process, the model's misclassification rate remains stable, while the adversarial attack success rate is significantly reduced. In addition, compared to other defense methods using noise, ITNA is shown to be much more efficient in terms of computing resources while maintaining the performance. It is easy and efficient to implement, as it only requires adding random noise to the raw waveform without additional AI modules during inference. We used ASVspoof2021 DF evaluation [28], In-the-Wild [29], and

DSD-Corpus [30] datasets, and the SSL_AASIST [31] and SSL_Conformer [32] models. Experimental results show that both models improve the original performance while also reducing the adversarial attack success rate.

The contributions of this paper are as follows:

- We introduce a lightweight, model-agnostic defense technique, ITNA, which can be seamlessly applied at inference time without retraining.
- We theoretically demonstrate the effectiveness of ITNA by analyzing the relationship between adversarial examples and the decision boundaries of an audio deepfake detection model.
- We experimentally demonstrate the effectiveness and efficiency of ITNA using three publicly available datasets (ASVspoof2021 DF evaluation(DF21), In-the-Wild(ITW), DSD-Corpus(DSD)) and two detection models (AASIST-based [33] and Conformer-based [34]).
- We demonstrate the effectiveness and efficiency of ITNA by comparing it to other noise-based defense systems and noise types.

II. BACKGROUND

This section outlines the technical background of our study. First, we introduce the general structure and operating principles of audio deepfake detection systems. We then discuss the threat and characteristics of adversarial attacks. Finally, we discuss the main defense methods proposed so far and their limitations.

A. AUDIO DEEFAKE DETECTION

Audio deepfake detection aims to determine whether a speech sample is genuine (bonafide) or synthesized (spoof). Detection systems typically consist of a front-end feature extractor and a back-end classifier. Early methods relied on handcrafted features [35], [36], combined with classifiers such as machine learning-based (e.g., GMM [37], SVM [38]), or deep learning-based networks (e.g., LCNN [16], ResNet [39], Res2Net [40], GAT [41]). However, their performance heavily relied on the quality of handcrafted features [42].

To address this, recent studies have increasingly adopted end-to-end architectures. These systems directly learn task-relevant representations from raw waveforms and optimize them jointly with the classifier. In particular, self-supervised learning-based models (e.g., Wav2Vec2.0 [43], HuBERT [44], WavLM [45]) have demonstrated strong performance in extracting meaningful audio representations and have become widely adopted in modern deepfake detection frameworks.

While end-to-end systems [30], [31], [32] achieve higher accuracy and better generalization, their reliance on deep neural networks also makes them more susceptible to adversarial attacks. This vulnerability underscores the importance of developing robust and efficient defense strategies to secure

audio deepfake detection in real-world scenarios. In the next subsection, we outline how adversarial attacks can exploit these vulnerabilities.

B. ADVERSARIAL ATTACK

Adversarial attacks exploit model vulnerabilities by introducing subtle, carefully designed perturbations to input data, causing incorrect predictions by the target model [21]. Attackers generally inject adversarial perturbation δ into input samples, such as the following formula:

$$x_{adv} = x + \delta \quad s.t. \quad \|\delta\|_p \leq \epsilon \quad (1)$$

where x_{adv} denotes the adversarial example, x is the original sample, $\|\cdot\|_p$ represents the L_p norm, and ϵ is the radius of the $L_p\epsilon$ -ball around x , which defines the allowed perturbations. Adversarial attacks on audio deepfake detection systems attempt to misclassify input audio samples into the opposite class by crossing the decision boundary. Adversarial attacks can be classified into white-box attacks and black-box attacks, depending on whether they have access to the target model information.

1) WHITE-BOX ATTACK

In a white-box attack setting, it is assumed that the attacker has full access to the internal information of the target model. This includes the model architecture, parameters, gradients, and knowledge of the training data. This level of access allows the attacker to compute the exact gradient of the loss function with respect to the input and generate highly effective adversarial examples with minimal perturbation. Several attack methods have been proposed in this setting [26], [46], [47], [48], [49], [50]. In this paper, we use the most representative white-box attack methods, Fast Gradient Sign Method (FGSM) [46], Projected Gradient Descent (PGD) [26].

FGSM is a single-step adversarial attack method that perturbs the input data in the direction of the sign of the gradient of the loss function. The perturbation is constrained to lie within the ϵ -ball. This approach is designed to move the input in the direction that increases the loss most rapidly, achieving both computational efficiency and strong attack effectiveness. The attack proceeds according to the following formula:

$$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x L(f(x), y)) \quad (2)$$

where L denotes the loss function, $f(x)$ denotes the model's prediction for input x , ∇ denotes the gradient, y denotes the true label of x , and $\text{sign}(\cdot)$ denotes the element-wise sign function.

Like FGSM, PGD perturbs the input in the direction of the gradient sign of the loss function, but it generates stronger adversarial examples through multiple iterations. At every iteration, the perturbation is projected into the allowed range ϵ -ball by clipping. This projection process ensures that the adversarial examples remain similar to the original input. The

attack proceeds according to the following formula:

$$x_{adv}^0 = x \quad (3)$$

$$x_{adv}^{t+1} = \Pi_{x+\delta} (x_{adv}^t + \alpha \cdot \text{sign}(\nabla_x L(f(x_{adv}^t), y))) \quad (4)$$

where x_{adv}^{t+1} denotes the adversarial example at iteration t , α is the step size of each gradient update, and $\Pi_{x+\delta}(\cdot)$ [26] denotes the projection operator that clips the result back into the ϵ -ball around the original input x .

In the context of audio deepfake detection, white-box attacks are particularly dangerous because they can precisely target the decision boundaries learned by the model [51]. Even a well-trained and seemingly robust model can be easily fooled when adversarial noise is crafted using full model knowledge. However, white-box attacks are often considered impractical in real-world applications because most deployed systems do not expose their internal structures or parameters. As a result, white-box attacks are mainly used in research settings as a benchmark to evaluate the worst-case robustness of detection models [52]. The following describes a black-box attack that attacks under restricted conditions.

2) BLACK-BOX ATTACK

In a black-box attack setting, the attacker does not have access to the internal structure or parameters of the target model. Instead, the attacker feeds inputs into the system and observes the corresponding outputs, such as class probabilities or predicted labels. This setting is considered more realistic in practical scenarios, where deployed systems often do not expose model internals for security and privacy reasons. Black-box attacks are generally classified into two categories: query-based attacks and transfer-based attacks. In this study, we focus on defending against transfer-based attacks.

Transfer-based attacks leverage the transferability property of adversarial examples. Adversarial inputs designed to fool one model can often fool another model with similar decision boundaries [53], [54]. In this approach, the attacker trains a substitute (or surrogate) model that mimics the target model's behavior as closely as possible, using publicly available data or query outputs [55]. Adversarial examples are then generated on the surrogate model and applied to the target system. The success of this attack depends on the degree of similarity between the surrogate and target models [56].

In audio deepfake detection, black-box attacks pose a substantial threat, as real-world systems are often accessed through embedded in applications where internal model details are inaccessible. This makes black-box attacks both feasible and dangerous. In particular, PGD-based attacks are known for their strong transferability [57]. To further improve the transferability of simpler attacks like FGSM, several enhanced variants have been proposed [58], [59]. As these transfer-based methods continue to evolve and exhibit increasingly generalized behavior, the need for defense mechanisms that do not rely on internal model knowledge becomes more critical.

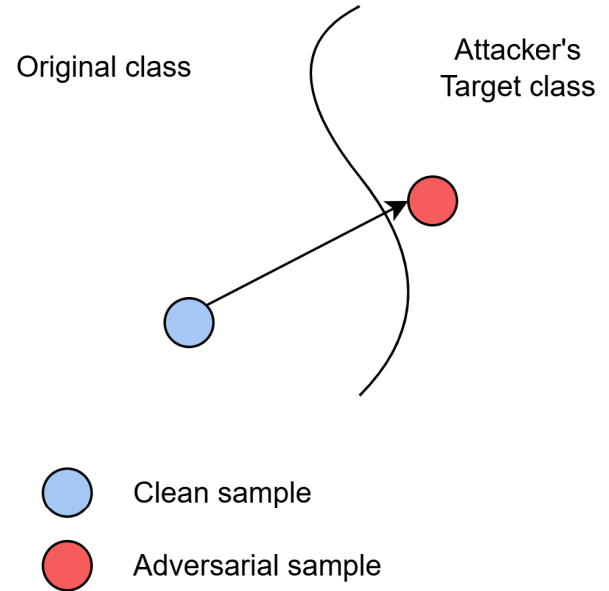


FIGURE 3. Changes in prediction result of target sample due to adversarial attack. The original class means true label of the input sample, and attacker's target class means the label that the attack causes the model to make a wrong prediction. For example, in audio deepfake detection, if the original class is bonafide, the target class becomes spoof.

3) ADVERSARIAL EXAMPLES AND DECISION BOUNDARY

Adversarial examples are generated by introducing minimal perturbations that push inputs across a model's decision boundary. FGSM and PGD, two representative white-box attacks, exploit gradients to maximize the model's loss while shifting the input toward a misclassified region under constrained distortion, as described in Equations (2) and (4). The perturbation is usually constrained by an ϵ -ball, as shown in Equation (1). Since the goal is to generate adversarial examples with minimal perturbation [60], [61], the adversarial examples barely cross the decision boundary or, in the case of iterative attacks like PGD, stop as soon as they cross the decision boundary [57]. Therefore, adversarial examples are generally distributed closer to the decision boundary than clean samples (non-attack samples) [62], as shown in Figure 3.

This observation is not limited to white-box attacks. Transfer-based black-box attacks exhibit similar spatial behavior in the input space. Previous research has shown that neural networks trained on the same task tend to learn similar or aligned decision boundaries [24]. This results in a shared decision subspace between independently trained models [53]. Recent research findings further demonstrate that adversarial examples generated by surrogate models designed to cross decision boundaries with minimal perturbation often lie close to the target's decision boundary [63].

C. ADVERSARIAL ATTACK DEFENSE

Adversarial attack defense is becoming an increasingly important research area in the development of robust deep learning models. In the context of audio deepfake detection,

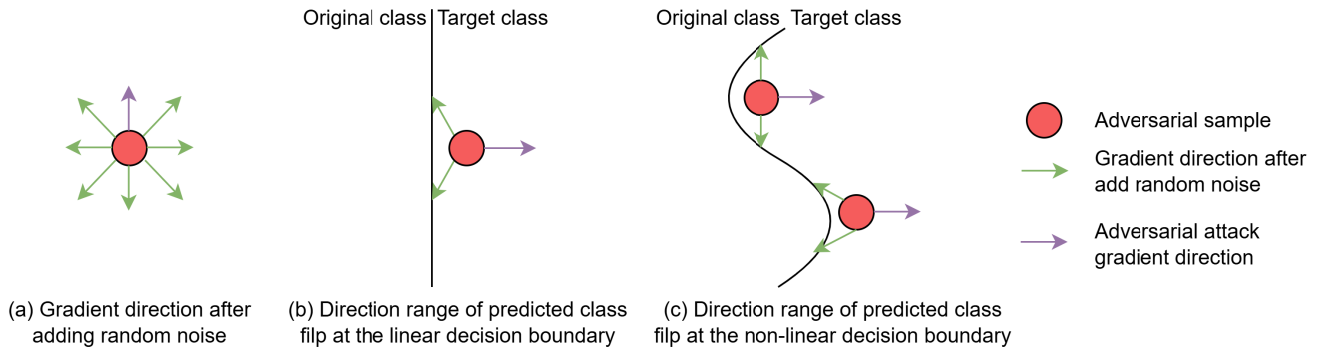


FIGURE 4. Adversarial attack gradient direction disturbed due to random noise.

where even imperceptible perturbations can cause critical misclassifications, robust defense strategies are crucial. In the audio domain, existing representative defense methods can be broadly classified into three main types: adversarial training, adversarial purification, and adversarial detection [64].

Adversarial training [26], [65] is one of the most straightforward and widely used defense techniques. It improves robustness by incorporating adversarial examples into the training process. While this approach is effective against known attacks, it often has high computational costs and limited generalization to unknown attack methods [66]. Collecting and training on new adversarial examples for every attack method is both difficult and inefficient, limiting its practicality as a general defense strategy.

Adversarial purification aims to remove or weaken adversarial perturbations from the input before it reaches the model. Common techniques include denoising [67], smoothing [68], re-synthesizing [69]. Although these methods can reduce the effectiveness of adversarial noise, they may also degrade the quality of inputs, thereby reducing model performance and user experience in practical applications. Recently, neural network-based denoising and resynthesis technologies [70] have been developed to minimize quality degradation. However, applying neural network-based pre-processing before feeding inputs into the detection model can introduce artifacts, potentially leading to misclassifications.

Adversarial detection involves training an auxiliary model or module to distinguish between clean and adversarial inputs [27]. Detection-based methods treat adversarial examples as out-of-distribution (OOD) or anomalous data and attempt to identify them using learned or handcrafted features [71], [72]. While these approaches can be effective against known attack types, they often struggle to generalize across different attacks and may exhibit high false positive rates, especially when adversarial examples closely resemble benign samples.

Although many defense methods [25], [73], [74], [75], [76] have been proposed to overcome these limitations, it should be noted that they were mainly conducted first in the image domain. Research focusing on audio-related tasks such as automatic speaker verification [77], [78], [79] and

TABLE 1. Comparison of retraining requirement, inference computation, and additional module requirement for ITNA and other defense methods. AT, AP, AD, and RS are Adversarial Training, Adversarial Purification, Adversarial Detection, and Randomized Smoothing, respectively. $F(x)$, $G_{AP}(x)$, $G_{AD}(x)$ represent the amount of computation when passing through the detection model, AP module, and AD module. n denotes the number of iterated inferences in RS.

Method	Retraining Requirement	Inference Computation	Additional module Requirement
AT	Require	$F(x)$	No
AP	Optional	$G(x) + F(x)$	Yes
AD	Optional	$G(x) + F(x)$	Yes
RS [25]	Optional	$n \times F(x)$	No
ITNA	No	$F(x)$	No

speech recognition [80], [81], [82] is on the rise. However, there is a relative lack of research in the field of audio deepfake detection [19]. Moreover, even among existing works in this domain, few have addressed the practical needs of lightweight, model-agnostic defenses that can be applied at inference time without retraining or modifying model architecture. These limitations highlight the need for lightweight, model-independent defenses that can operate at inference time without compromising detection performance.

III. METHODOLOGY

In this study, we propose ITNA, a lightweight yet theoretically grounded defense mechanism against adversarial attacks in audio deepfake detection. ITNA operates by adding Gaussian noise into the raw waveform during the inference process to perturb the adversarially generated input vector and cause it to cross the decision boundary. The key to ITNA is that adversarial perturbations are generally minimal [61] and adversarial examples are located near the decision boundary [62]. On the one hand, random noise can shift such inputs back into their correct classification region, especially when adversarial examples lie near the decision boundary.

Table 1 compares ITNA and other defense methods in terms of the retraining requirement, inference computation, and the additional module requirement. Adversarial training only passes the detection model once during inference, so it

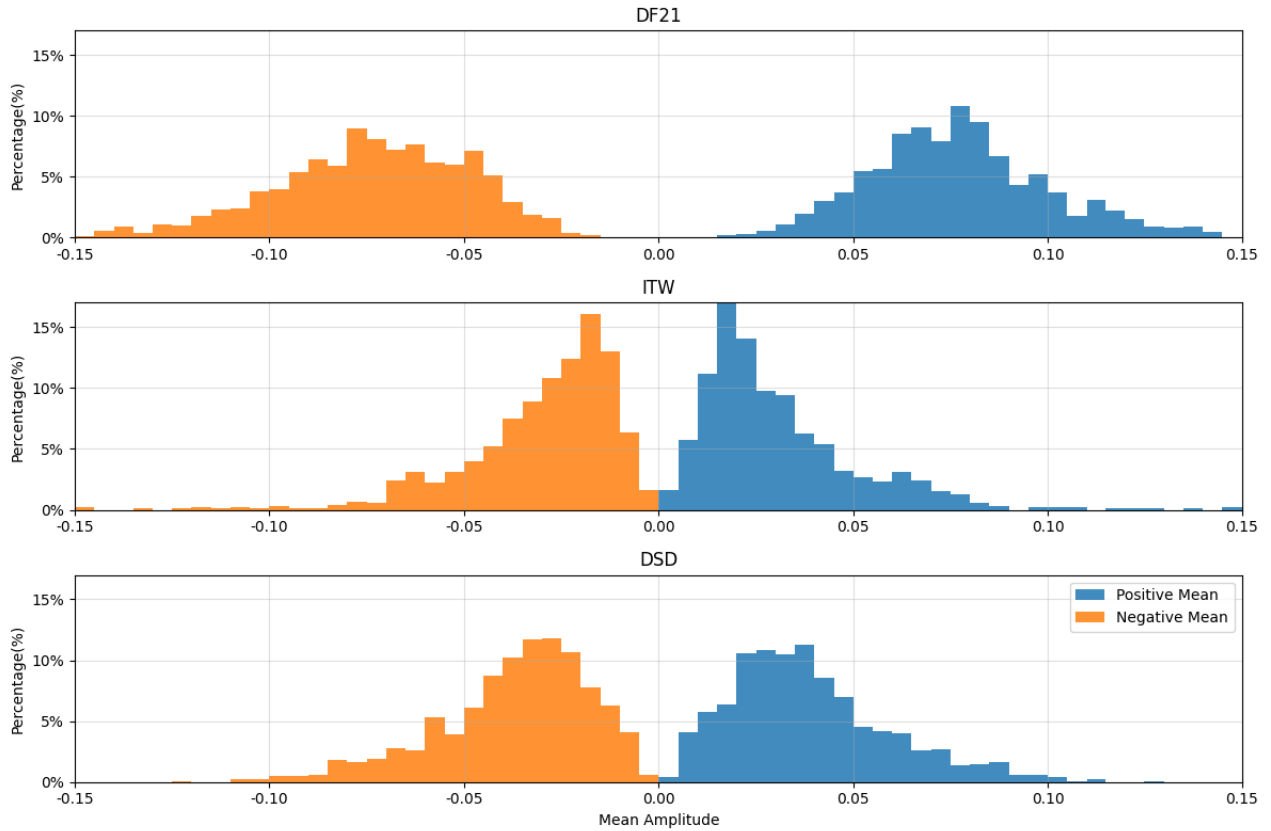


FIGURE 5. Distribution of positive and negative mean amplitude value across datasets. The x-axis represents the range of positive and negative mean amplitude values in the waveform. The y-axis represents the percentage (%) of samples in each amplitude range.

computes $F(x)$, but the retraining process requires high computational cost. Adversarial purification and adversarial detection require purification and detection modules during inference. Pre-trained modules can be used, but using the modules incurs additional computation $G(x)$ and inference latency. Randomized smoothing can improve effectiveness through retraining using Gaussian noise as an augmentation, but retraining is optional. However, the noise must be sampled n times for each input and repeated inference is performed. Increasing n for higher certification significantly increases the computational cost to $n \times F(x)$. On the other hand, ITNA does not require any retraining, simply performs $F(x)$ at inference time, and does not require additional modules. This comparison shows that ITNA is more efficient than other methods. In the following sections, we demonstrate through theoretical analysis and empirical evaluation that ITNA effectively suppresses attack success rates across multiple datasets and model architectures. At the same time, it maintains or even improves generalization performance in some environments.

A. EFFECT OF NOISE ADDITION

When random noise is added to a sample, the loss value of the sample in Equation (6) changes.

$$x_{noise} = x + \eta \text{ and } \eta \sim N(0, \sigma^2 I) \quad (5)$$

$$L(f(x), y) \quad (6)$$

where σ^2 denotes the variance of the Gaussian distribution N and I is the identity matrix, so each component of η is independently drawn from $N(0, \sigma^2 I)$. Therefore, the gradient due to adversarial perturbation can be very different after noise is added, as shown in (a) of Figure 4. In other words, adding random noise can shift the sample's position within the input space. Figures 4 (b) and 4 (c) illustrate how an adversarial example perturbed by random noise can be shifted back into the correct class region under both linear and non-linear decision boundaries. In the linear case, the example can return to the original class within a specific angular range. A similar phenomenon is observed in the non-linear case. Modern models use complex model structures to classify samples by leveraging various features of the data. However, as the model structure becomes more complex, it forms more complex decision boundaries [83]. Therefore, the prediction of the model for adversarial examples distributed near the decision boundary can be more sensitively changed by random noise than for clean samples.

When a sample x is located at a distance D from the decision boundary and Gaussian noise with standard deviation σ is added, the probability P that the sample passes

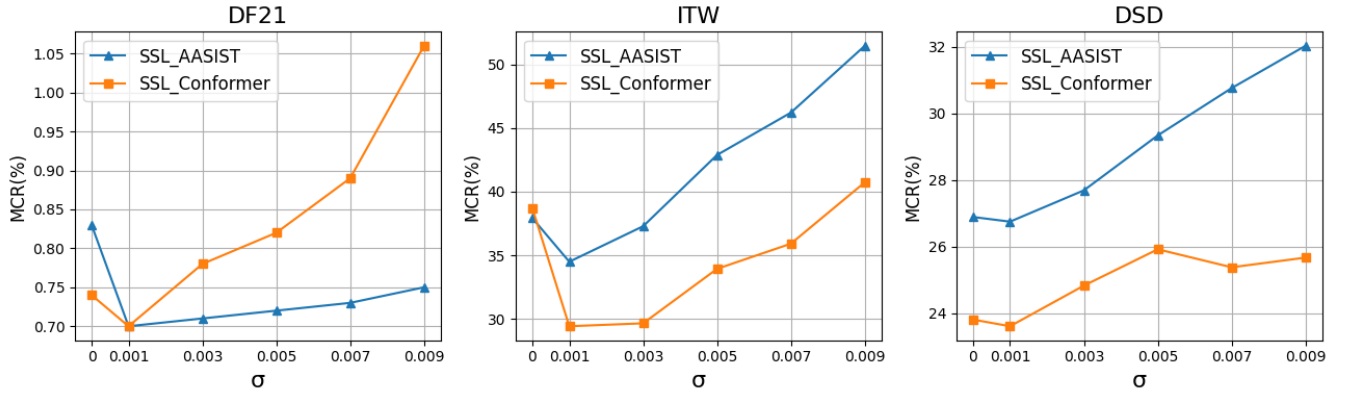


FIGURE 6. Trend of changes in misclassification rates (MCR) of the two models for each dataset as the σ value increases.

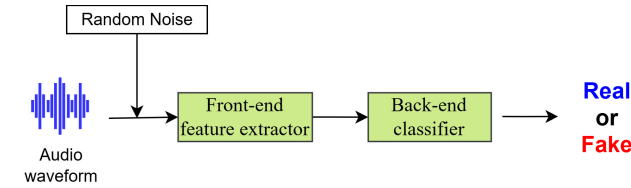


FIGURE 7. ITNA workflow: Adding random noise during the inference phase.

the decision boundary is given by the following formula:

$$P = \Phi\left(-\frac{D}{\sigma}\right) \quad (7)$$

where Φ denotes the cumulative distribution function of the standard normal distribution. According to [84], Equation (7) holds exactly for linear boundaries, such as those in Figure 4 (b). It can also be approximated for complex neural network models with nonlinear decision boundaries, such as those in Figure 4 (c). Reconstructing this formula for distance D gives:

$$D = -\sigma \Phi^{-1}(P) \quad (8)$$

Adversarial examples are typically closer to the decision boundary than clean samples because adversarial attacks aim to minimize the perturbation needed to misclassify the target. Consequently, under the same noise size, the expected distance from the decision boundary satisfies Equation (9).

$$\mathbb{E}[D_{clean+\eta}] > \mathbb{E}[D_{adv+\eta}] \quad (9)$$

where $\mathbb{E}$ is the expected value and η is the random noise defined in Equation (5). Thus, ITNA can help shift adversarial examples back to their correct class regions while preserving detection performance. However, if σ is too small, it may not sufficiently alter the adversarial input; if it is too large, it may excessively perturb clean inputs, leading to incorrect detections. Ultimately, the value of σ must find a balance that sufficiently disrupts adversarial examples while not harming clean inputs. However, this balance varies depending on factors such as the data distribution, model architecture, and the strength of the adversarial attack. Therefore, finding the

TABLE 2. Total numbers of original and adversarial datasets.

Dataset	DF21	ITW	DSD
Bonafide	14,869	19,963	46,330
SpooF	519,059	11,816	46,055
Adversarial	38,630	39,731	40,000

TABLE 3. Number of successful adversarial attack examples for the target model for each dataset.

Dataset	SSL_AASIST		SSL_Conformer	
	FGSM	PGD	FGSM	PGD
DF21	9,270	10,000	9,360	10,000
ITW	9,988	9,908	9,835	10,000
DSD	10,000	10,000	10,000	10,000

optimal σ is difficult, so we determined the range of σ through experiments.

B. IMPLEMENTATION DETAILS

As shown in Figure 7, ITNA adds random noise to the raw waveform before feeding the input audio to the model. The noise is sampled from a Gaussian distribution and has the same length as the waveform vector. The noise size must be carefully determined because adding too much noise can seriously damage the sound quality. To this end, when choosing a Gaussian distribution, the mean is fixed to 0, and σ is adaptively changed so that the randomly selected noise does not increase or decrease overall. In this paper, instead of optimizing σ for a specific model or dataset, we aim to evaluate the overall effectiveness of ITNA. For this purpose, we choose three representative σ values of 0.001, 0.003, and 0.005. These values are selected based on the size of the waveform data extracted using the librosa library,¹ which normalizes the waveform values in the range -1.0 to 1.0 .

To further clarify the selection rationale, we analyzed 1000 audio samples from ASVspoof 2021 DF evaluation

¹<https://librosa.org/>

TABLE 4. Misclassification rate(MCR) change according to σ value. Bold indicates improved MCR values after adding noise.

MCR(%)	DF21				ITW				DSD			
σ value	0	0.001	0.003	0.005	0	0.001	0.003	0.005	0	0.001	0.003	0.005
SSL_AASIST	0.83	0.70	0.71	0.72	37.92	34.49	37.28	42.88	26.89	26.75	27.69	29.34
SSL_Conformer	0.74	0.70	0.78	0.82	38.70	29.42	29.65	33.93	23.81	23.61	24.83	25.92

TABLE 5. Attack success rate(ASR) change according to σ value. Attack the model next to the attack method with an adversarial example generated through the target model. If the target model and the attack model are the same, it is a white-box attack, and if they are different, it is a transfer-based black-box attack.

ASR(%)		DF21				ITW				DSD			
σ value		0	0.001	0.003	0.005	0	0.001	0.003	0.005	0	0.001	0.003	0.005
Target Model →		SSL_AASIST											
FGSM	SSL_AASIST	92.70	71.58	49.11	37.45	99.88	80.95	62.69	55.62	100	94.57	73.21	59.65
	SSL_Conformer	67.84	63.98	53.05	45.41	85.64	82.93	73.54	68.10	87.01	84.01	71.77	62.55
PGD	SSL_AASIST	100	79.71	50.16	28.33	99.08	90.55	47.23	9.07	100	90.16	74.61	66.69
	SSL_Conformer	67.77	64.57	45.67	33.99	77.18	72.83	41.89	21.93	84.28	82.41	77.03	73.76
Target Model →		SSL_Conformer											
FGSM	SSL_AASIST	41.55	39.05	30.48	24.95	51.65	38.79	18.53	10.75	81.56	81.75	81.26	77.86
	SSL_Conformer	93.60	79.64	60.01	49.32	98.35	87.06	59.98	44.31	100	99.09	96.22	91.52
PGD	SSL_AASIST	41.89	36.69	18.76	10.36	74.84	64.47	22.48	3.71	76.30	72.05	56.41	43.31
	SSL_Conformer	100	77.42	41.56	25.29	100	88.73	45.39	20.41	100	95.85	77.02	62.32

set (DF21) [28], In-The-Wild (ITW) [29], and DSD-Corpus (DSD) [30]. For each sample, we computed the mean positive and negative amplitude values of the waveform elements. As shown in Figure 5, most of the positive and negative mean amplitude values across all datasets are concentrated within the range of approximately -0.1 to 0.1. Based on this observation, we selected small σ values (0.001, 0.003, and 0.005), corresponding to roughly 1% to 5% of the typical amplitude scale. This range was chosen to ensure that the injected noise perturbs the input sufficiently to disrupt adversarial examples near the decision boundary, while still preserving the overall acoustic characteristics of the original waveform as much as possible. As shown in Figure 6, the misclassification rate increases as the value of σ increases. As the value exceeds 0.005, audio quality and performance deteriorate significantly. Therefore, we empirically limited the maximum σ to 0.005.

IV. EVALUATIONS

A. EXPERIMENTAL SETUP

We evaluate the proposed method from two perspectives: (1) its impact on the general performance of the model, and (2) its effectiveness in defending against adversarial examples.

For this purpose, we use SSL_AASIST [31] and SSL_Conformer [32] models, which show state-of-the-art performance in audio deepfake detection. Both model was trained using ASVspoof2019 [85] training and development data. For evaluation, the ASVspoof2021 DF evaluation set is used as the in-domain dataset. To assess out-of-domain generalization, we use the In-The-Wild dataset,

which includes SNS data, and the DSD-Corpus dataset, which contains various state-of-the-art synthesizer samples.

We use FGSM and PGD as adversarial attack methods. In each attack, ϵ -ball was set to 0.01, and in PGD, the step size α was set to 0.0005 and the number of steps was set to 20. In addition, PGD stopped immediately when the attack was successful in a step. Since the ITW dataset contains approximately 10,000 spoof samples, we generate adversarial examples by attacking 10,000 spoof samples from each dataset using both FGSM and PGD, applied to both models. The specific dataset configuration is shown in Tables 3 and 4. We also evaluate defense performance against transfer-based black-box attacks by applying adversarial examples generated through white-box attacks on one model to the other model.

We used the misclassification rate (MCR) and the attack success rate (ASR) as metrics to evaluate the change in the performance of the model through ITNA and the attack success rate against adversarial examples. MCR reflects how often the model makes incorrect predictions due to added noise, while ASR measures the change in the success rate of adversarial attacks after applying ITNA. MCR and ASR are calculated by the following formulas:

$$MCR = \frac{\text{False Positives} + \text{False Negatives}}{\text{Total Samples}} \times 100 \quad (10)$$

$$ASR = \frac{\text{Successful Attack Samples}}{\text{Total Attack Samples}} \times 100 \quad (11)$$

For both MCR and ASR, the lower the value, the better the performance.

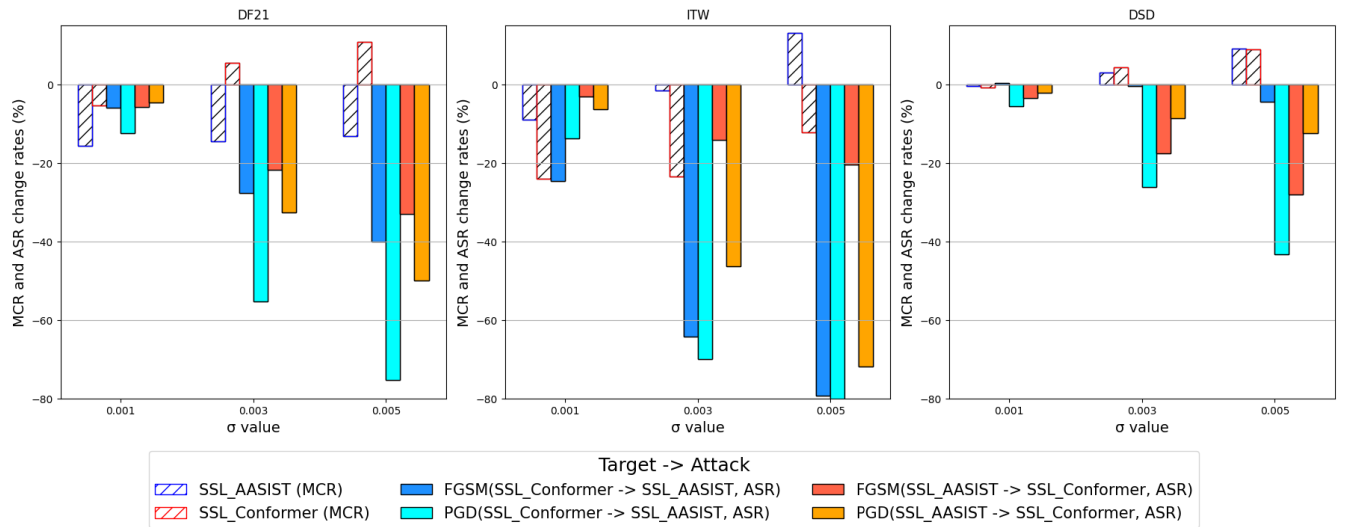


FIGURE 8. Misclassification Rate (MCR) and Attack Success Rate (ASR) change rate in transfer-based attack.

B. EXPERIMENT RESULTS

1) EFFECT OF NOISE ADDITION FOR MCR

Table 4 shows that adding a small amount of noise decreases the MCR. This can be attributed to the fact that the model was trained using noise based data augmentation such as rawboost [86]. Therefore, the addition of Gaussian noise acts as data augmentation, recognizing it closer to the trained pattern of the model [87]. However, as the noise size increases, the audio quality and acoustic characteristics degrade, leading to a gradual increase in MCR. The MCR increase of the in-domain dataset DF21 is small, while the MCR increases of the out-of-domain datasets ITW and DSD are large. This can be interpreted as showing more robust results for DF21 data because the ASVspoof 2019 training data and DF21 data refer to the same VCTK dataset [28]. On the other hand, ITW already contains noise, and DSD is out-of-domain data, so it is less robust to external noise. In Table 4, when the σ value is 0.001 and 0.003, the MCR value decreases or increases by up to about 1%p (percentage point), indicating that ITNA does not significantly affect the general performance of the model.

2) EFFECT OF NOISE ADDITION FOR ASR

Table 5 shows the ASR changes for FGSM and PGD attack samples. When the noise size is too small, its effect on adversarial examples is limited, resulting in minimal defensive benefit. On the other hand, when the noise size is large, the ASR decreases a lot in all cases. In this case, the effect of moving the adversarial example to the original class is more pronounced. However, there is also an effect where the attacked sample, which was intended to be recognized as bonafide, is classified as spoof. This occurs because the audio quality or acoustic features are degraded due to significant noise. This is consistent with the increase in the MCR value as the noise size increases, as shown in

Table 4. Notably, the ASR for PGD attacks on the ITW dataset decreases significantly. This is likely because the combination of existing noise and added perturbations leads to substantial distortion, causing the model to classify the input as spoof. Table 5 shows that the larger the noise size, the more the attack success rate is reduced on average. However, the goal of this study is not to analyze the effects of noise on MCR and ASR independently, but to evaluate their combined impact under the same noise conditions.

3) ANALYZING THE MCR AND ASR CHANGE RATE

To evaluate the trade-off between MCR and ASR across different noise sizes, we compute the change rate based on the original and changed values using the following formula:

$$\text{change rate} = \frac{\text{Change value} - \text{Original value}}{\text{Original value}} \times 100 \quad (12)$$

The more negative this value, the better the general performance and defense performance.

Based on Table 5, we evaluate the effectiveness of ITNA by analyzing the change rates of MCR and ASR under transfer-based attacks, where ASR reduction is generally smaller compared to white-box attacks. Figure 8 shows that for the DF21 and ITW datasets, both MCR and ASR are improved, or when the σ value is 0.003 or 0.005, the ASR improvement rate is much higher than the MCR performance degradation rate, which enables effective defense. For the DSD dataset, when the σ value is 0.001, both MCR and ASR show minimal changes. However, for σ is 0.003 and 0.005, ASR improvement surpasses MCR degradation, except in the case of transfer-based FGSM attacks on the SSL-AASIST model (represented by the dark blue bar). This exception indicates that unlike ITW and DF21, DSD does not have severe base noise and the effect of adding noise is relatively

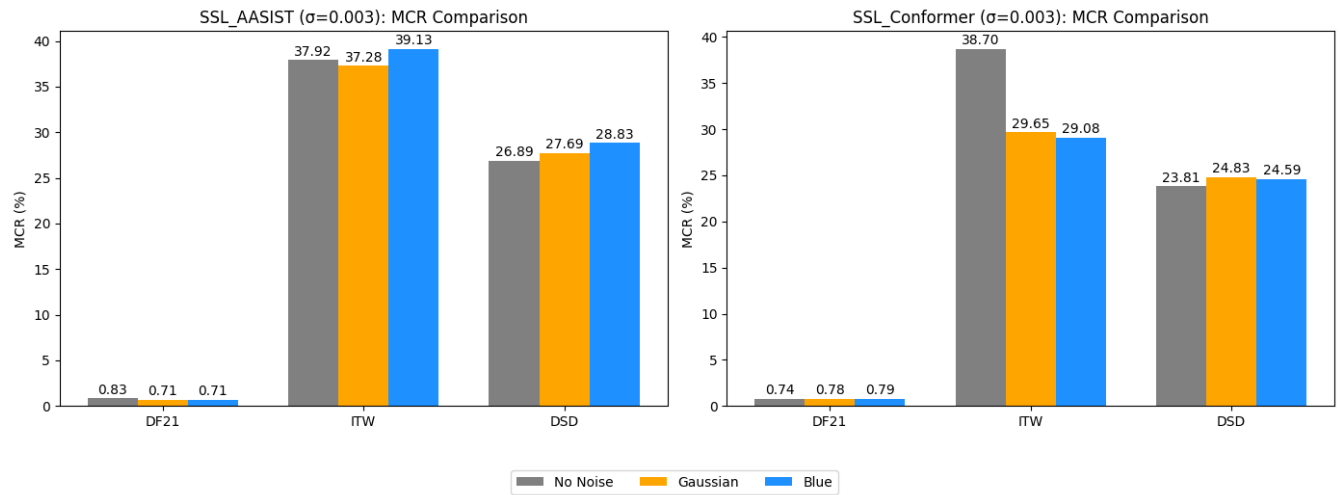


FIGURE 9. Misclassification Rate (MCR) Comparison of Blue Noise vs. No Noise and Gaussian Noise ($\sigma = 0.003$).

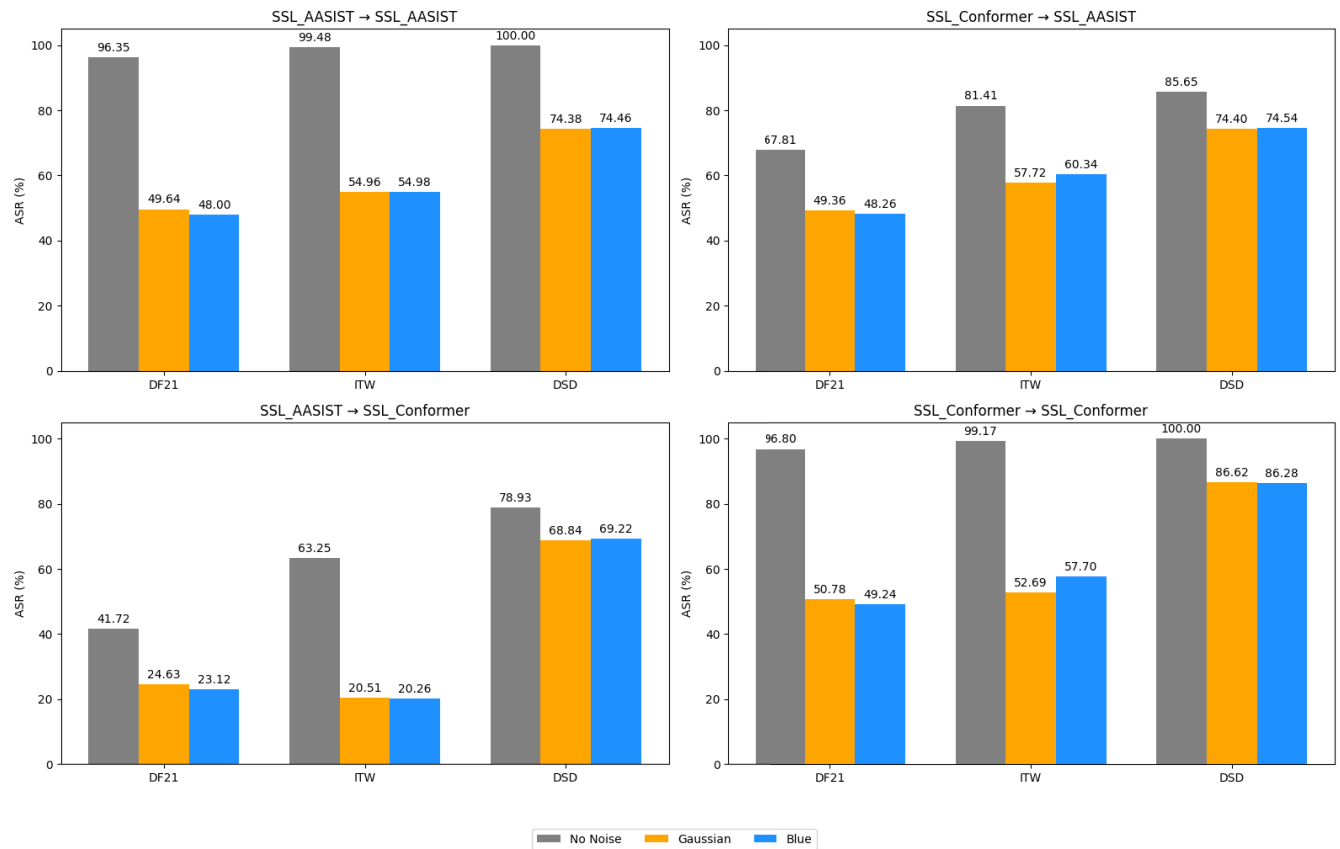


FIGURE 10. Attack Success Rate (ASR) Comparison of Blue Noise vs. No Noise and Gaussian Noise ($\sigma = 0.003$). Target model -> Attack model.

weak. DSD is an out-of-domain dataset and is based on a transfer-based attack, meaning that the adversarial examples were not generated through an optimized attack. Therefore, the same size of noise does not have as significant an effect as it does on other datasets. However, it can reduce ASR in other cases, although not as much as DF21 and ITW.

In this experiment, we show that ITNA is generally effective even without using the optimal σ value. In particular, when the σ value is 0.003, the average MCR and ASR values for the three datasets of the two models decreased by 4.48% and 31.90%, respectively. These results imply that ITNA can improve the robustness against adversarial attack examples

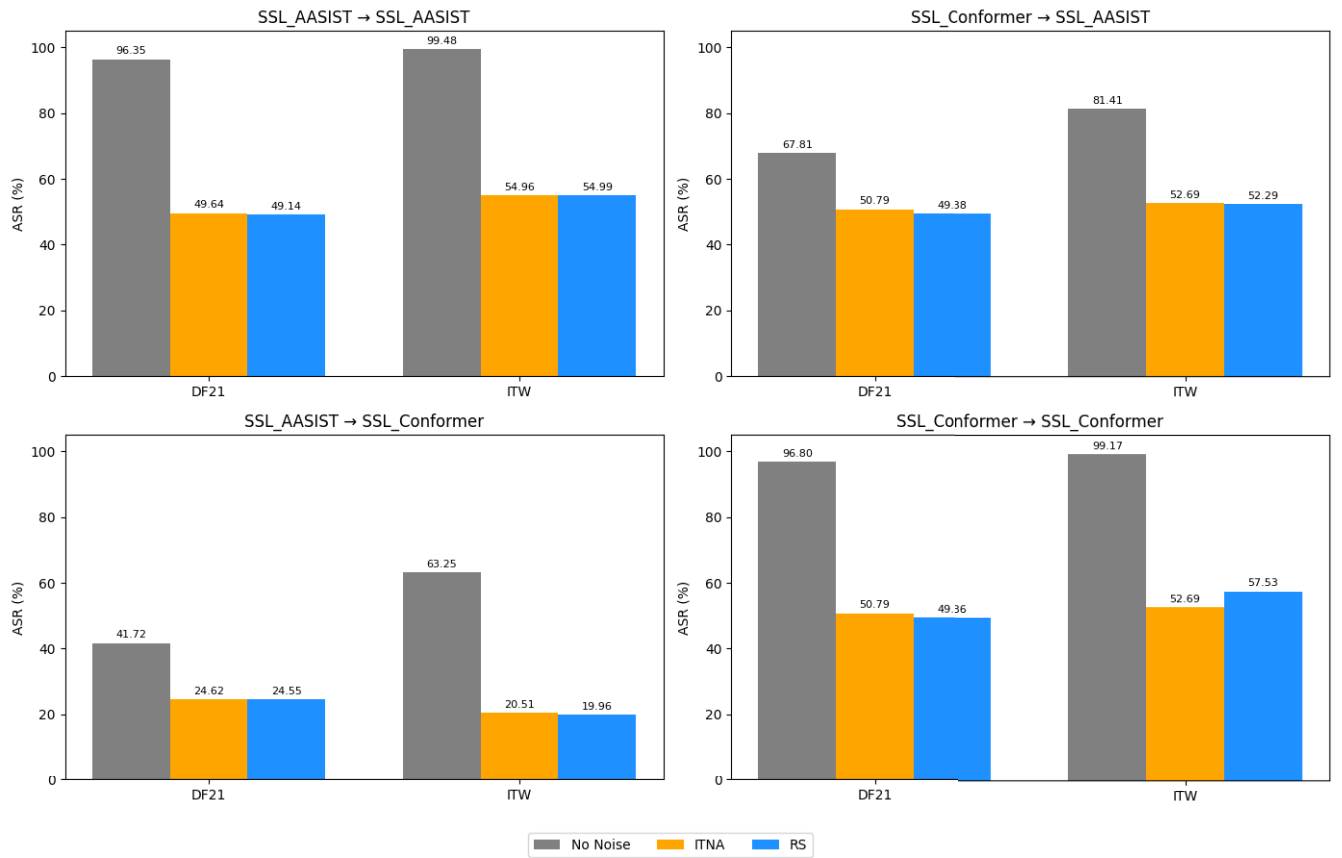


FIGURE 11. Attack Success Rate (ASR) Comparison of Randomized smoothing (RS) vs. No Noise and Gaussian Noise ($\sigma = 0.003$). Target model → Attack model.

without significantly affecting the general performance of the model. If an appropriate value for σ is found through hyperparameter search, it can be used as a more effective defense method.

C. COMPARATIVE ANALYSIS

To further assess the generality and effectiveness of ITNA, we conducted comparative experiments involving alternative noise type and other noise-based defense method.

1) EVALUATION OF DIFFERENT NOISE TYPE

While our proposed method primarily uses Gaussian noise, we also explored whether other noise types with distinct spectral characteristics could provide similar or complementary defense effects. Prior research suggests that adversarial perturbations in the audio domain often concentrate in high-frequency regions [88], [89], [90]. Based on this observation, we investigated whether adding noise that emphasizes high-frequency components could more effectively disrupt adversarial patterns.

To this end, we conducted comparative experiments using blue noise [91], which emphasizes higher frequency components more than Gaussian noise. Blue noise has a power spectral density proportional to frequency ($\propto f$). We generate blue noise of the same length as the input

waveform and scale its amplitude using a σ value as a scaling factor, identical to that used in Gaussian noise. The scaled blue noise was then added to the raw waveform using the same ITNA procedure.

Referring to the results in Tables 4 and 5, σ was set to 0.003, which effectively reduces ASR while maintaining MCR. As shown in Figure 9, the addition of blue noise generally shows similar changes in MCR as when adding Gaussian noise. Figure 10 shows the average adversarial attack success rate with blue noise added compared to Gaussian. The results consider ASR for both methods as a whole, without separating FGSM and PGD. In some cases, blue noise slightly outperforms Gaussian noise at certain σ values, which may be due to the high-frequency enhancement effect of blue noise. In other cases, the opposite is true. However, overall, it shows a similar level of ASR reduction as the addition of Gaussian noise. These results suggest that while blue noise may provide advantages in certain situations due to its frequency characteristics, it generally exhibits the same protective tendencies as Gaussian noise.

2) COMPARISON WITH OTHER NOISE BASED METHOD

To further evaluate the effectiveness and practicality of ITNA, we compare it with a widely studied noise-based defense method known as randomized smoothing [25]. Randomized

smoothing (RS) adds random noise to the input and classifies the sample based on the majority vote over multiple noisy inferences. While it has been shown to provide certifiable robustness in classification tasks, especially in the image domain, its application to audio tasks is relatively limited.

In this experiment, we implement a baseline RS method for audio deepfake detection by averaging predictions for multiple noise inputs for DF21 (in domain) and ITW (out-of-domain). Specifically, we decide σ as 0.003 for the same reason as the blue noise experiment, and determine the final classification based on majority voting for 100 noise samples.

As shown in Figure 11, the defense performance of ITNA is comparable to that of RS across various datasets, models, and noise sizes. This result implies that Gaussian noise with a limited range can shift input samples in a limited range, and within the shift range, the probability of shifting to the original class is guaranteed with a single noise addition. Therefore, unlike RS, which requires multiple inferences per input, ITNA achieves similar robustness with a single inference step, making it significantly more efficient for real-time applications.

V. DISCUSSION

The experimental results demonstrate that ITNA effectively reduces the adversarial attack success rate while maintaining model performance across different datasets and architectures. ITNA exhibits strong generalization, even in out-of-domain scenarios, and achieves performance comparable to randomized smoothing, while requiring significantly fewer computational resources. Its simplicity and efficiency make it highly suitable for real-time applications. Due to its model-agnostic nature and inference-time operability, ITNA is highly compatible with other defense mechanisms. It can potentially be combined with other methods to further enhance robustness against a wider range of attacks. In particular, attacks against adversarial trained models may enable broader defenses than the trained attack types. These characteristics make ITNA a promising foundation for practical and lightweight adversarial defense in audio deepfake detection. However, the effectiveness of ITNA depends on the chosen noise size, and currently, there is no established method for determining the optimal noise intensity across different attack scenarios or datasets. Developing adaptive noise scheduling strategies remains an open research challenge, and investigating adaptive σ tuning will be explored as part of our future work for improved generalization.

VI. CONCLUSION

In this paper, we proposed ITNA, a simple yet effective defense mechanism against adversarial attacks on audio deepfake detection systems. ITNA works by adding Gaussian noise during inference stage, requiring no model retraining or architectural modification. Through extensive evaluations on multiple datasets and state-of-the-art models, our proposed method reduces the adversarial attack success rate by an

average of 31.90% while reducing the misclassification rate by 4.48%. Compared to techniques such as randomized smoothing, ITNA achieves similar defense performance with significantly lower computational cost and latency. Due to its model-agnostic nature and ease of integration, ITNA is highly suitable for real-time, low-resource, or embedded applications. In future research, we plan to study an adaptive algorithm that selects the optimal σ depending on model behavior or input characteristics to maximize the effect of ITNA.

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